

Investigating Mechanisms of Nocturnal Ozone in Northwest Indiana and Chicagoland

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Abstract

In 2025, Valparaiso University (VU) joined the National Ozone Garden Network. This nationwide program uses ozone-sensitive plants to monitor the effects of ozone pollution alongside an in situ ozone monitor. During the summer of 2025, ozone concentrations exceeded the U.S. Environmental Protection Agency (EPA) National Ambient Air Quality Standard of 70 ppb only nine times, with the highest concentration occurring on 18 September, 2025. This peak occurred during a regional ozone episode from 14-21 September, 2025, when multiple Air Quality Action Days were issued across Northwest Indiana and the Chicago metropolitan area. To investigate the meteorological conditions and transport processes responsible for elevated ground-level ozone concentrations during this event, surface ozone observations from the VU Ozone Garden, EPA monitoring stations, and the Community Research on Climate and Urban Science (CROCUS) sensor network were combined with meteorological observations from Automated Surface Observation System (ASOS) stations, and synoptic weather analyses. By comparing ozone concentrations, wind patterns, and regional weather conditions, the influence of ozone transport from the Chicago metropolitan area and the role of lake breeze circulations in producing elevated ground-level ozone concentrations in Valparaiso can be determined. Understanding the factors that contributed to this high-ozone event will improve knowledge of ozone episodes in Northwest Indiana and support future air quality forecasting and public health awareness.

Introduction

Clean air in the troposphere is composed of oxygen (O_2) and nitrogen (N_2) molecules. Nitrogen oxides (NO_x) and volatile organic compounds (VOCs) are air pollutants emitted by the combustion of fossil fuels. The VOCs absorb oxygen atoms during chemical reactions, which then react with nitrogen monoxide (NO) to form nitrogen dioxide (NO_2). In sunlight, NO_2 reacts, losing an oxygen atom (O), which then bonds to an oxygen molecule to form ozone (O_3) (UCAR, 2026). Unlike primary pollutants that are emitted directly into the atmosphere, ozone formation depends strongly on both precursor emissions and meteorological conditions. Consequently, episodes of elevated ground-level ozone values often occur during periods of increased solar radiation, warmer air temperatures, weak atmospheric mixing, and persistent transport patterns that allow ozone to accumulate (An et al., 2024). Exposure to elevated ozone concentrations has been associated with numerous adverse health effects, including respiratory irritation, decreased lung function, and increased hospital admissions (EPA, 2026). Ozone also negatively impacts agricultural productivity and ecosystem health by reducing average crop yields by an average of 3.5% for corn and 6.1% for soybeans, causing an estimated economic loss of 2.6 billion dollars in the United States (Pei et al., 2024), making accurate characterization of ozone episodes important for both public health and environmental management (Heath and Grulke, 2019).

Under limited cloud coverage, surface ozone concentrations peak in the afternoon as solar radiation reaches its maximum and decrease rapidly after sunset (An et al., 2024). However, elevated ozone concentrations are occasionally observed at night, a phenomenon known as nocturnal ozone. These nighttime enhancements have been linked to several atmospheric processes, including vertical mixing of ozone-rich air from aloft, transport by

nocturnal low-level jets, residual layer entrainment, and long-range advection of ozone produced during the daytime (He et al., 2023). Identifying the mechanisms responsible for nocturnal ozone events is essential for improving air quality forecasting and crop resiliency, and understanding the evolution of regional pollution episodes, particularly in areas influenced by both urban emissions and complex boundary-layer dynamics.

The occurrence of nocturnal ozone is of interest in the Great Lakes region, where the interaction of synoptic-scale weather systems, lake-breeze circulations, and regional transport creates a highly dynamic atmospheric environment. The Chicago Metropolitan area is a substantial source of ozone precursors, while prevailing winds often transport ozone-rich air masses into northwestern Indiana (Gray and Finster, 2015). Under favorable meteorological conditions, ozone transported from the Chicago region can remain aloft in the residual layer during the evening, then be mixed downward or advected into rural communities overnight (Harris and Kotamarthi, 2005). As a result, locations such as Valparaiso, IN (Figure 1) may experience elevated nighttime ozone concentrations despite minimal local photochemical production. Identifying the meteorological mechanisms responsible for these nocturnal ozone events is essential for improving air quality forecasting, refining our understanding of regional ozone transport, and assessing human and ecological exposure to ozone outside of the traditional afternoon pollution maximum.

During 14-21 September, 2025, prolonged air quality alert days caused by ozone affected northeastern Illinois and northwestern Indiana, prompting multiple Air Quality Action Days and producing the highest ozone concentrations observed at the Valparaiso University Ozone Garden during the 2025 growing season (ozone season 1 March through 31 October) (EPA, 2026). While elevated daytime ozone concentrations during regional pollution episodes are well documented,

the persistence of enhanced ozone concentrations after sunset in Valparaiso warrants the investigation of atmospheric transport and boundary-layer processes. A better understanding of these interactions improves our knowledge of ozone variability in northwest Indiana.

Data and Methods

Surface ozone measurements were collected from the Valparaiso University Ozone Garden (VU) during the 14-21 September 2025 air quality episode, with particular emphasis on the nocturnal ozone event observed on 18 September. To provide regional context, ozone observations from the U.S. Environmental Protection Agency (EPA) Air Quality System (AQS) monitoring network and the CROCUS sensor network were obtained for sites across the Chicago metropolitan area and northwest Indiana. CROCUS sites were chosen based on availability of data for the period (Figure 1). Total ozone column datasets came from the European Centre for Medium-Range Weather Forecasts (ECMWF) Reanalysis 5th Generation (ERA5) data (Hersbach et al. 2023). CROCUS hourly ozone concentrations were converted from micrograms per cubic meter ($\mu\text{g}/\text{m}^3$) to parts per billion (ppb), and the total ozone column from ERA5 was converted from kilograms per square meter (kg/m^2) to Dobson Units (DU) by a factor of 46620.

Meteorological conditions were determined from observations at Automated Surface Observing System (ASOS) stations, CROCUS weather observations, local meteorological tower measurements, and ERA5 data. Variables analyzed included 2-m air temperature, 10-m wind speed and direction, total ozone column, and surface pressure. Hourly ERA5 reanalysis data were used to generate regional maps of total column ozone, 2-m air temperature, and 10-m wind speed and direction. These maps were used to characterize synoptic-scale meteorological conditions and evaluate regional transport patterns associated with the 14-21 September 2025 ozone episode. These analyses were used to determine whether periods of persistent southerly

flow, weak nighttime winds (0-6 knots), and favorable meteorological conditions were conducive to pollutant transport. Tropospheric Emissions: Monitoring of Pollution (TEMPO) satellite observations of tropospheric NO₂ were also analyzed to examine the spatial distribution of ozone precursor emissions and evaluate how Lake Michigan lake-breeze circulations influenced pollutant transport across the Chicago metropolitan area and northwest Indiana.

To investigate the origin of air masses arriving at Valparaiso during nocturnal ozone episodes, backward trajectories were computed using the NOAA Hybrid Single-Particle Lagrangian Integrated Trajectory (HYSPLIT) model. Trajectories at 500 millibars (mb) were initialized over Valparaiso multiple times on 18 September, 2025, and traced backward in time to identify potential source regions contributing to elevated ozone concentrations. The trajectory analysis was used to assess whether transported air masses originated from the Chicago metropolitan area or other upwind regions. Additionally, archived soundings were used to assess boundary layer conditions and review wind plots through the Wyoming Upper Air Data Archive, along with reanalyzed soundings created using ERA5 data at specific sites in the region.

Time-series analyses were performed to compare average hourly ozone concentrations among Valparaiso, EPA monitoring sites, and CROCUS stations. Ozone observations were averaged over each hour to produce hourly mean ozone concentrations for each monitoring site. These hourly averages were used for all subsequent time-series comparisons among Valparaiso, EPA monitoring sites, and CROCUS stations. To evaluate ozone exposure following EPA methodology, 8-hour rolling average ozone concentrations were calculated by averaging each set of eight consecutive hourly observations. The maximum daily 8-hour average (MDA8) was then identified for each day and compared among monitoring sites. Particular attention was given to the timing, magnitude, and duration of daytime maxima and secondary nocturnal peaks. Solar

insolation data were evaluated alongside ozone observations to distinguish between daytime photochemical production and nighttime transport processes.

The combined meteorological, trajectory, and ozone analyses were used to assess whether persistent nighttime ozone concentrations resulted primarily from regional transport and downward mixing from the residual layer, a stable layer of air above the nocturnal boundary layer that often retains ozone-rich air from the previous afternoon, rather than local photochemical production. Downward mixing was measured indirectly by combining elevated nocturnal ozone concentrations with meteorological conditions and trajectory analyses consistent with entrainment. In this process, air from one layer of the atmosphere is mixed into another layer, with ozone-filled air from the residual layer entering the nocturnal boundary layer.

Results

During 14-21 September 2025, elevated ozone concentrations were observed throughout northwestern Illinois and northwestern Indiana, with the highest concentration at VU, reaching 74 ppb on the night of 18 September 2025. Average hourly ozone concentrations at the Valparaiso (Valpo) EPA sensor and at available CROCUS sites showed the typical daytime increase associated with photochemical production, rising after sunrise and peaking as solar radiation reaches its maximum around solar noon (1230 LT). Throughout the week, select EPA and CROCUS sites displayed large diurnal amplitudes, with hourly concentrations decreasing to nearly 0 ppb at night and reaching 78-88 ppb during the day across the region. VU, however, showed no distinct peak, as ozone concentrations continued to rise after sunrise until reaching a peak near 2330 LT (Figure 3). Additionally, VU's average hourly ozone concentrations showed a narrower range than at the other sites, remaining around 20-30 ppb after the peak rather than fully depleting. Elevated ground-level ozone concentrations were recorded across the region

during 14-21 September, as evidenced by the issuance of Air Quality Action Days for northeastern Illinois and northwest Indiana for each day.

At VU, the steady rise in ozone continued past sunset, ultimately reaching the nocturnal peak at 2330 LT. This pattern occurred every night during the period, with VU showing a single peak at this time rather than a pronounced daytime peak followed by a secondary peak after sunset, like the Valpo EPA and CROCUS sites (Figure 2). Nighttime concentrations at VU exceeded those measured at several nearby CROCUS sites, indicating localized differences in nocturnal ozone behavior rather than a regionally uniform enhancement. This persistence is consistent with regional transport of ozone-rich air masses and nighttime boundary-layer processes. However, the relative contributions of horizontal transport and vertical mixing cannot be determined from the available observations. The Valpo EPA sensor recorded a secondary, smaller peak that occurred around the same time as VU's on every night during the period. In contrast, only a few CROCUS sites observed a secondary peak on certain nights, suggesting that the responsible mechanism was not consistent across the network. Additionally, VU's 8-hour average ozone concentration did not exceed the EPA threshold of 70 ppb. In contrast, the Chicago Water Filtration Plant (CHISWFP) and the Valpo EPA each exceeded it on at least one day (Figure 1, Figure 3).

Synoptic conditions during the period included above-average September temperatures and predominantly cloud-free skies over northern Illinois, which promoted strong daytime photochemical production. Temperatures reached 31.7°C, which is 12°C warmer than average September temperatures for the area. TEMPO NO₂ observations on 18 September showed higher precursor concentrations over the Chicago metropolitan area than in surrounding rural areas, while ERA5 reanalysis indicated enhanced total column ozone extending across northern Illinois

and northwest Indiana. HYSPLIT backward trajectories initialized over Valparaiso at 1800 UTC (1300 LT) on 18 September indicated that air masses originated from multiple directions, suggesting a complex regional transport pattern associated with the prevailing synoptic system and suggesting that regional transport contributed to elevated ozone over northwest Indiana ahead of the nighttime event.

On 18 September, ERA5 reanalysis, which combines observations with numerical weather model output, indicated 0-6 kt winds and a localized area of higher total column ozone (295-300 DU) just southeast of VU between 2300 LT and 0300 LT on 19 September. This area weakened by 0600 LT-1200 LT, and ozone levels across the region became more consistent. Throughout the afternoon, winds increased over northern Illinois, and ozone levels decreased as the lake breeze progressed through the western suburbs of Chicago, while remaining elevated over Northwest Indiana. The largest differences occurred in areas near the strongest winds by 1700-1800 LT.

Discussion

The 14-21 September 2025 episode demonstrates the strong influence of synoptic conditions on regional ozone variability in northwest Indiana. Although ozone formation depends on the presence of NO_x, VOCs, and sunlight, the timing and magnitude of ozone concentrations are likely influenced by atmospheric transport, temperature, and other processes (UCAR 2024). During this event, EPA and CROCUS monitoring sites generally exhibited the expected diurnal cycle, with ozone concentrations peaking around solar noon. VU, however, consistently experienced increasing ozone concentrations into the night, resulting in a peak around 2330 LT. The persistence of this pattern throughout the study period indicates that

processes other than daytime photochemical production influenced ozone concentrations in the area.

Since ozone production via photochemical reactions essentially ceases after sunset, elevated nighttime ozone concentrations are generally attributed to atmospheric transport and boundary-layer dynamics rather than to new chemical production. Previous studies, such as An et al. (2024), have identified several mechanisms responsible for nocturnal ozone, including transport of ozone-rich air masses, residual-layer entrainment, and vertical mixing. Although this study examined total ozone column evolution using ERA5 analysis, this does not directly resolve the vertical distribution within the atmosphere. The combined trajectory, meteorological, and ozone analyses support an important role for regional transport, but the extent to which the residual-layer entrainment and vertical mixing contributed to the observed nocturnal ozone concentrations cannot be definitively determined. Future observations using ozonesondes or an ozone lidar would better characterize the vertical ozone structure and clarify the importance of these processes.

Meteorological observations collected during the study support the importance of regional transport. Above-average temperatures approaching 31.7°C in Northern Illinois, clear skies, and solar radiation favored efficient daytime photochemical ozone production throughout the region. Similar meteorological conditions have been identified as major contributors to elevated ozone episodes, as warmer temperatures accelerate photochemical reactions, while stagnant conditions allow ozone to accumulate (An et al., 2024). TEMPO observations on 18 September also showed elevated NO₂ concentrations over the Chicago metropolitan area, indicating precursor emissions during the event. HYSPLIT backward trajectories demonstrated that air masses arriving in Valpo originated from multiple directions, reflecting the regional

circulation associated with the synoptic pattern rather than transport from a single source region. Synoptic analyses indicated a broad ridge of high pressure over the eastern United States during 14-21 September 2025, producing persistent southwesterly flow across the Midwest. This circulation favored transport of ozone and its precursors from the Chicago metropolitan area into northwest Indiana. Together, these observations suggest that elevated ozone measured at VU resulted from regional-scale transport processes interacting with favorable meteorological conditions.

Lake Michigan likely also contributed to the evolution of the event, though not through direct surface-level transport. The afternoon lake breeze shifted cooler air inland across the Chicago shoreline. TEMPO and ERA5 data showed elevated ozone extending across northern Illinois and northwest Indiana during this period, with some studies finding that lake breezes are known to trap and concentrate pollutants near the shore before letting them spread further inland later in the day (Harris and Kotamarthi, 2005). However, 10-m wind observations and surface wind roses indicated that surface winds were predominantly from the southwest during this event, rejecting the theory that the surface air over Valpo originated from Chicago. This finding suggests that any influence of Chicago-area emissions on the nocturnal ozone event in Valpo was unlikely to have occurred through advection of ozone-rich air from the lake. A more probable explanation is that ozone produced near Chicago during the day may have been mixed into the residual layer, transported horizontally aloft, and later entrained into the nocturnal boundary layer as the boundary-layer structure evolved overnight. Although this mechanism is consistent with previous studies of nocturnal ozone transport, it remains a hypothesis in this study because vertical profile observations were not available

The occurrence of persistent ozone has important implications for both public health and ecosystem health. Exposure to elevated ground-level ozone has been associated with numerous health effects (EPA, 2026). Ozone also damages vegetation by reducing photosynthesis and impairing plant growth (Heath and Grulke, 2019). Although the VU garden did not exceed the EPA 8-hour National Ambient Air Quality Standard during this period, prolonged nighttime ozone concentrations increase cumulative ozone exposure for both humans and ozone-sensitive plants. Since the National Ozone Garden Network is specifically designed to evaluate ozone impacts on vegetation, the persistent nocturnal ozone observed in this study highlights the value of combining biological observations with continuous atmospheric measurements to better understand ecosystem exposure.

Conclusion

This study examined the 14-21 September 2025 ozone episode in northwest Indiana and northeastern Illinois, with particular focus on the nocturnal ozone peaks recorded at Valparaiso University. While EPA and CROCUS monitoring sites across the region generally followed the expected diurnal ozone cycle, with concentrations peaking in the afternoon and declining after sunset, VU consistently found that there was an ozone increase throughout the day that continued after sunset, reaching a nocturnal peak around 2330 LT through the period, including the garden's highest recording of 74 ppb on 18 September 2025 at 2300 LT. The Valpo EPA sensor showed a smaller secondary peak around the same time as VU's, alongside various CROCUS sites, indicating the presence of a localized feature.

Above-average temperatures, cloud-free skies, and elevated NO₂ precursor concentrations over the Chicago metropolitan area supported strong daytime ozone production across the region. They enhanced the total ozone column, extending from northern Illinois into

northwest Indiana. Although local surface winds were predominantly southwesterly, HYSPLIT backward trajectories indicated that the air masses reaching Valparaiso followed multiple transport pathways, reflecting the broader regional circulation rather than transport from a single source region. This finding suggests that elevated nocturnal ozone at VU was more consistent with regional transport than with simple direct surface-level transport from Chicago. The mechanism by which ozone reached the surface, however, could not be determined from the available observations.

Overall, this ozone air quality episode demonstrates that nocturnal ozone behavior can vary considerably over short distances even within a single regional pollution event. Persistent nighttime ozone concentrations at a given site cannot be assumed to follow the same drivers as the broader regional ozone signal. Although the VU and Valpo EPA sites are located in close proximity, they exhibited different nocturnal ozone behavior. The cause of these site-to-site differences could not be determined with the available observations but may reflect local-scale variations in land use, surface roughness, boundary-layer structure, or measurement siting. Additional observations would be needed to identify the processes responsible. There is a large need for future research based on these results. Since surface wind observations could not confirm a direct transport pathway from Chicago, deploying ozonesondes to characterize vertical ozone profiles would help determine whether ozone-rich air aloft could contribute to the nocturnal maximum at VU. Further investigation into the role of Lake Michigan and lake-breeze circulations in regional ozone transport is also warranted, given the limited spatial and temporal reach of the lake breeze relative to VU's distance from the shoreline. Expanding this analysis to additional Midwestern monitoring sites would help determine how frequently similar nocturnal ozone events occur beyond Valparaiso, and further investigation of the Ozone Garden using new

sensors is planned for the end of summer 2026. It will provide continued opportunity to refine understanding of this recurring nocturnal enhancement and improve regional air quality forecasting in northwest Indiana.

Figures:

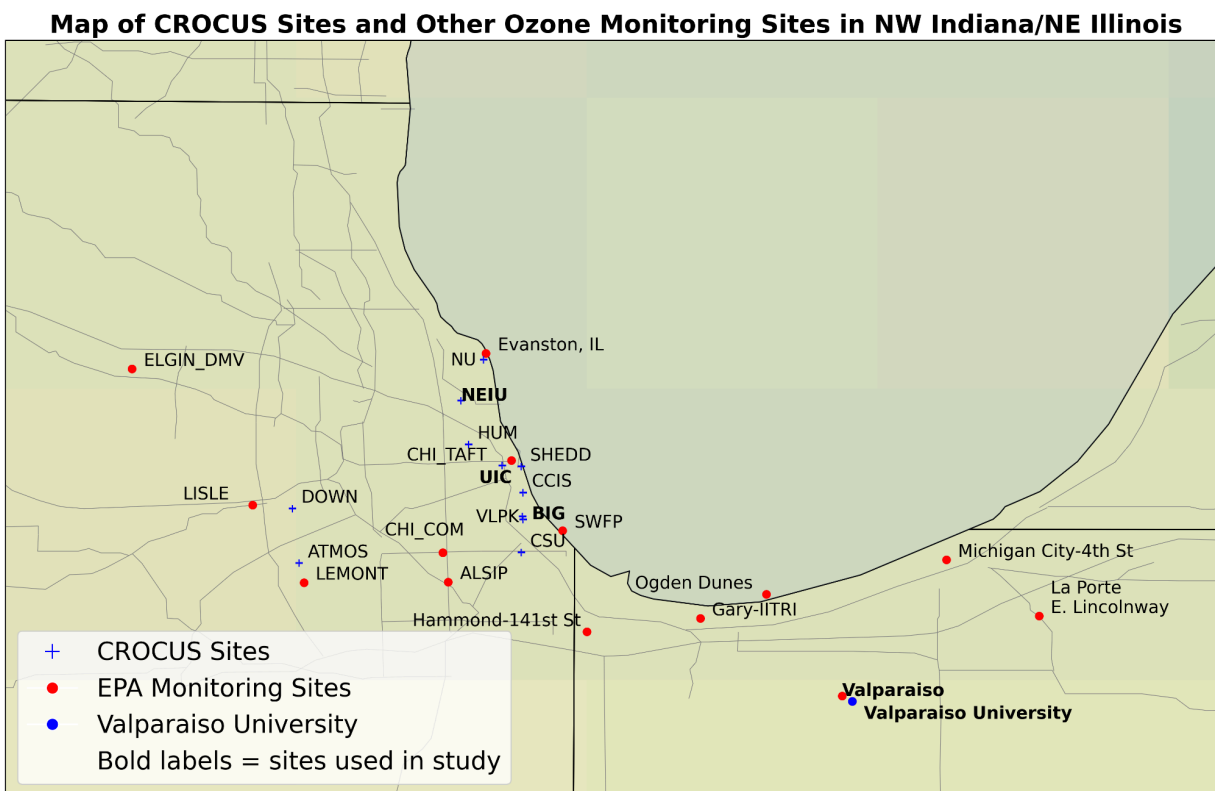


Figure 1. Map of locations of potential air quality monitoring sites. Red dots are EPA sensor sites, blue crosses are CROCUS sites, and the blue dot is Valparaiso University; bolded sites are sites used in the study.

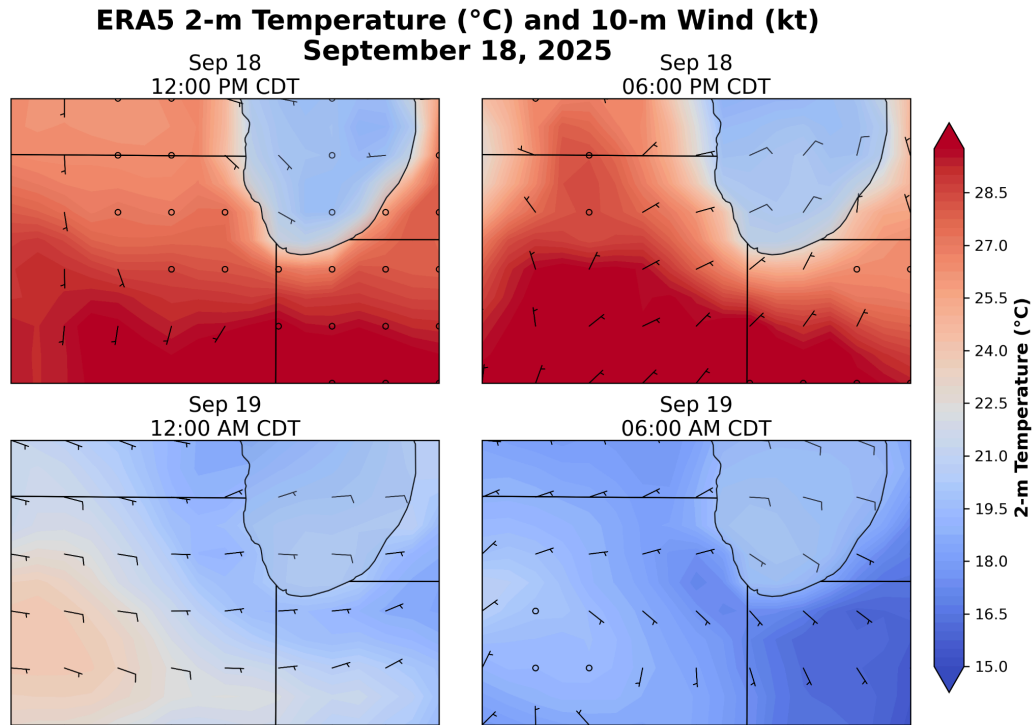


Figure 2. Evolution of 2-m temperature (°C) and 10-m wind (kts) on 18 September, 2025 from ERA5.

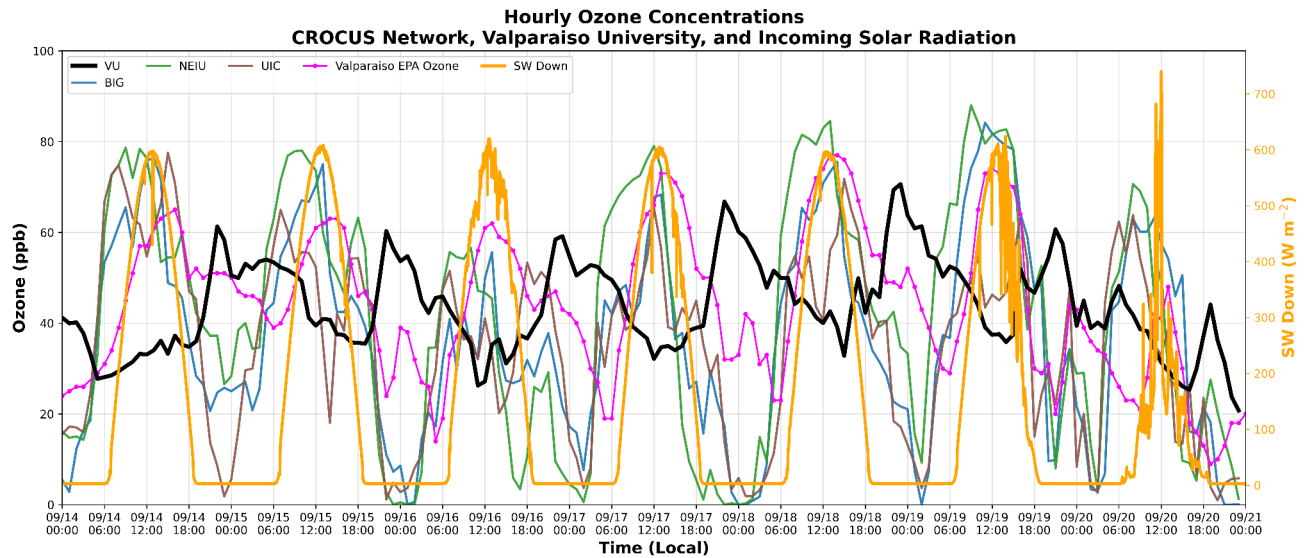


Figure 3. Time series showing when O₃ peaks occurred at each site along with shortwave radiation (right y-axis) measured in watts per square meter (W/m²) from the VU tower. VU is Valparaiso University (black), NEIU is North Eastern Illinois University (green), UIC is University of Illinois Chicago (magenta), and BIG is West Woodlawn “Blacks in Green” (blue).

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